

SWA project

In this course we studied many different topics related to software architecture. We implemented many different aspects with spring cloud. Suppose I am your project lead and you and all your classmates work in the same project where we implement a microservice architecture on premise using spring cloud (Spring cloud gateway, Spring cloud load balancer, ...) and the tools we used in this course (Consul, Keycloak, Kafka, ...). I want to know what would be involved when we would build the same microservice architecture in AWS cloud.

Build a minimal microservice architecture (only 2, 3 or 4 microservices) that run in AWS cloud and make use of the following aspects:

- Registry
- Load balancing
- API gateway
- Tracing
- Logging
- Resilience: circuit breaker
- Config server
- Security
- REST & Messaging

The goal of the project is **not** to build a full fledged microservice architecture. The goal is to understand what changes when we implement our spring cloud based architecture in AWS cloud, and how we need to change our existing architecture.

Do some research into this, write one or more demo's and give a presentation on this for your team members. It is your responsibility that your team members get a good overview of developing our existing microservice architecture into AWS cloud, how it works and how the architecture looks like in the cloud.

- Create a demo that shows the different aspects of your system running in the cloud.
- Explain the important points of your demo and your architecture decisions.
- What are the strong and weak points?
- How would a cloud architecture be different compared to the architectures we studied in this course?

The microservices that you build should use both **REST calls** between each other as well as **asynchronous events using messaging**. Your microservices can use a simple cloud database of your choice. This project is not about complex database persistence.

In the same way, I do not care how you implemented your microservices. Make them as simple as possible and only implement the code you really need for your demo implementation.

Your microservice architecture should make use of the following supporting services:

- Registry
- Load balancing
- API gateway
- Tracing
- Logging
- Resilience: circuit breaker
- Config server
- Security

It is important that in your presentation you show and explain the **different options** that are available, explain their **advantages and disadvantages**, and explain which option you choose and **why you choose this option**.

To summarize, this project has 3 goals:

1. Learn the **different options** that are available to implement a microservice architecture in spring cloud and understand their characteristics.
2. Implement a **demo** microservice architecture that you can demo during the presentation.
3. Communicate all your findings clearly in an interesting 30 minute **presentation**. The goal is that the class **learns** about cloud architecture by attending your presentation.

The SWA project is a research project you do in groups of 2.

For this project you have **4 ½ days**

- Friday: **½ day (afternoon)**
- Saturday: **1 day**
- Monday : **1 day (We all meet Monday at 11:00 in our classroom)**
- Tuesday: **1 day**
- Wednesday: **1 day (We all meet Wednesday at 11:00 in our classroom)**
- **Thursday: presentations.**

What to hand in?

- The demo code/configuration you used.
- The slides used during the presentation

The project needs to be submitted in the assignment section in Sakai.

**The due date for submitting the deliverables is Thursday, November 20 at 9:45 AM.
(Material submitted after Thursday at 9:45 AM will lose points)**

In class presentation

- You give your presentation to the class on **Thursday November 20**. Your presentation should be around **30 minutes** long. Make sure both members of your group present about the same time length (15 minutes per person). Make sure your presentation does **not take longer than 30 minutes**.
- Assume that all people that listen to your presentation have the knowledge at the level of what you learned in this course. So your presentation should build on top of that knowledge. **Do not repeat anything we know already** because we learned it in this course. The goal of this project is that we learn something new from each other. For example, do not explain what a circuit breaker or an API gateway is.
- Keep your presentation interesting for your audience. (Show demo's, write on the whiteboard, talk clearly) Do not read from your slides. If you want to explain your code, do it in an interesting way so that your audience does not fall asleep.
- **Capture all important research findings in your slides**. Use as much **diagrams** and pictures as possible. Do not use slides with a lot of text.
- **Speak slowly, loud and clearly**

The grading of this project is based on:

- The quality of your research. (Demo, did you research the relevant aspects, conclusions)
- The quality of your in-class presentation